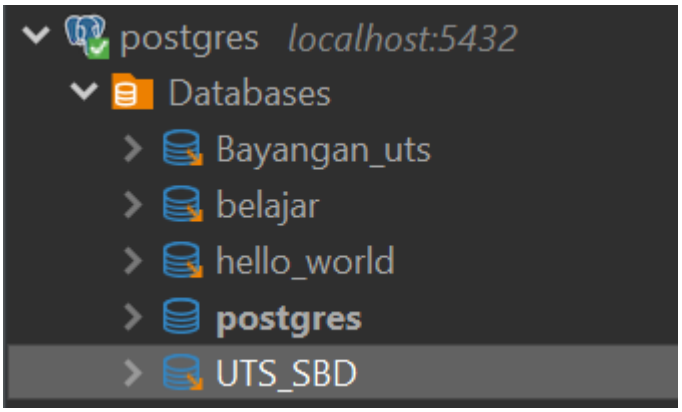
UTS SBD

Nama : Haesam Abdullah
NIM : 1251420101

DOKUMENTASI HASIL PENGERJAAN

1. Buatkan Data Base Terlebih dahulu



2. Lalu kita buatkan masing masing tabel kolom yang di perlukan dalam pengerjaan soal berikut;
   - Buat kolom tabel hotel

```
CREATE TABLE Hotel (
    hotelNo VARCHAR(10) NOT NULL,
    hotelName VARCHAR(50) NOT NULL,
    city VARCHAR(50) NOT NULL,
    PRIMARY KEY (hotelNo)
);
```

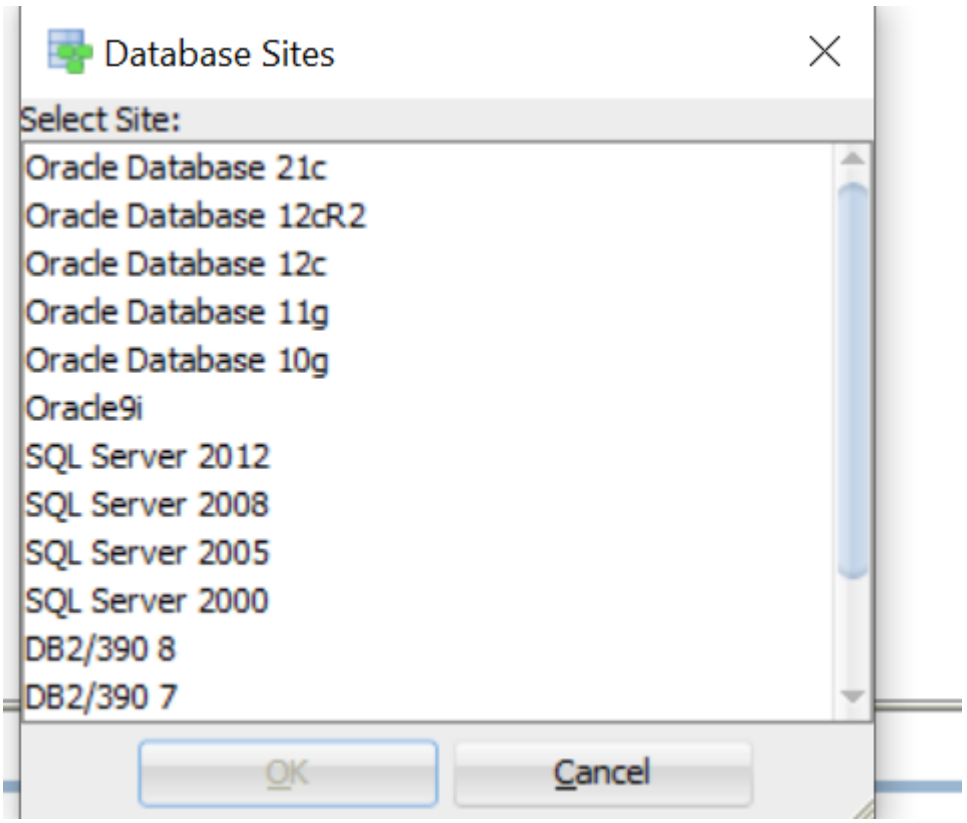
   - Buat kolom tabel guest, booking , room

```sql
CREATE TABLE Room (
    roomNo INT NOT NULL,
    hotelNo VARCHAR(10) NOT NULL,
    type VARCHAR(10) DEFAULT 'Single' NOT NULL,
    price DECIMAL(5, 2) NOT NULL,
    PRIMARY KEY (roomNo, hotelNo),
    FOREIGN KEY (hotelNo) REFERENCES Hotel(hotelNo),
    CONSTRAINT chk_room_type CHECK (type IN ('Single', 'Double', 'Family')),
    CONSTRAINT chk_room_price CHECK (price BETWEEN 10 AND 100),
    CONSTRAINT chk_room_no CHECK (roomNo BETWEEN 1 AND 100)
);

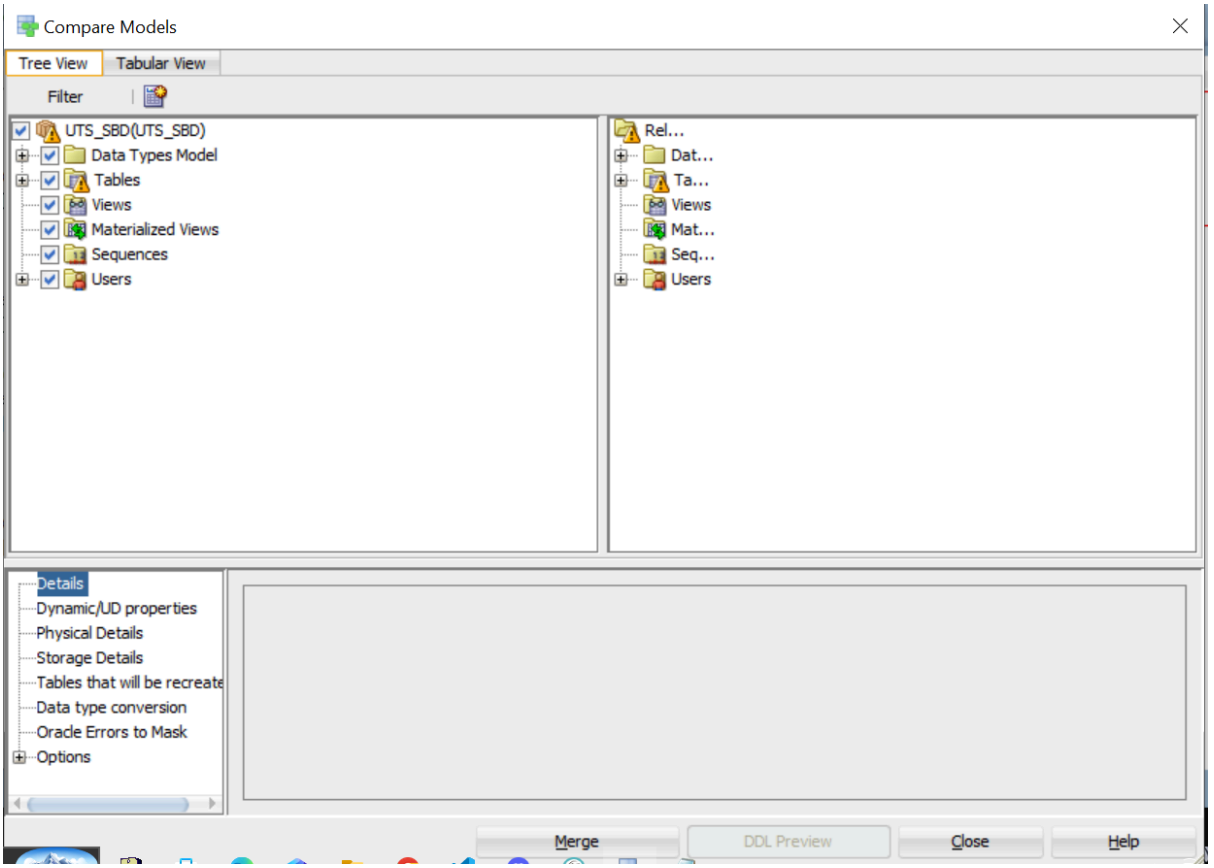
CREATE TABLE Guest (
    guestNo VARCHAR(10) NOT NULL,
    guestName VARCHAR(50) NOT NULL,
    guestAddress VARCHAR(100) NOT NULL,
    PRIMARY KEY (guestNo)
);

CREATE TABLE Booking (
    hotelNo VARCHAR(10) NOT NULL,
    guestNo VARCHAR(10) NOT NULL,
    dateFrom DATE NOT NULL,
    dateTo DATE NOT NULL,
    roomNo INT NOT NULL,
    PRIMARY KEY (hotelNo, guestNo, dateFrom),
    FOREIGN KEY (hotelNo) REFERENCES Hotel(hotelNo),
    FOREIGN KEY (guestNo) REFERENCES Guest(guestNo),
    FOREIGN KEY (roomNo, hotelNo) REFERENCES Room(roomNo, hotelNo)
);
```

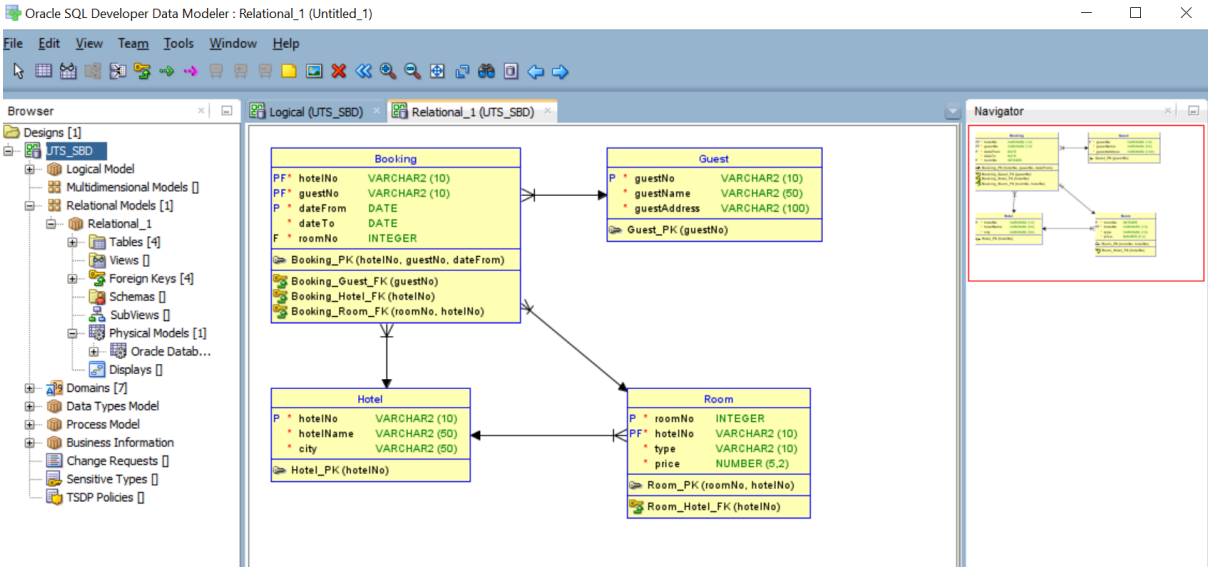
3. Setelah di buatkan file database hotel tersebut maka kita eksport ke dalam file .txt atau .sql untuk kita lakukan import ke Oracle SQL developer data model
4. Simpan query di notepad sebagai `.sql`
5. Import ke Data Modeler.
6. Buat physical model dengan target PostgreSQL.
7. Generate DDL dari physical model
8. Kita masuk ke bagian Oracle SQL developer data model
   Setelah import file .sql tdi kita pilih ini menggunakan oracle databse 12c



9. Lalu kita Merge pada compire model



10. Maka akan menghasilkan tabel relasiny atau ERD



MENGGUNAKAN TOOLS ORACLE

1. Buka notepad untuk membuat catatan sql
2. Siapkan masing masing tabel yg harus kita buatkan
    - Berikut isi query yg perlu ada

```sql
CREATE TABLE Branch (
    branchNo VARCHAR(10) NOT NULL,
    mgrStaffNo VARCHAR(10),
    street VARCHAR(50),
    city VARCHAR(30),
    postcode VARCHAR(15),
    CONSTRAINT Branch_PK PRIMARY KEY (branchNo)
);

-- 2. Tabel Staff
CREATE TABLE Staff (
    staffNo VARCHAR(10) NOT NULL,
    supervisorStaffNo VARCHAR(10),
    branchNo VARCHAR(10) NOT NULL,
    fName VARCHAR(30),
    lName VARCHAR(30),
    position VARCHAR(20),
    sex CHAR(1),
    DOB DATE,
    salary NUMERIC(10, 2),
    CONSTRAINT Staff_PK PRIMARY KEY (staffNo),
    CONSTRAINT Staff_Supervisor_FK FOREIGN KEY (supervisorStaffNo) REFERENCES Staff (staffNo),
    CONSTRAINT Staff_Branch_FK FOREIGN KEY (branchNo) REFERENCES Branch (branchNo)
);

-- Menambahkan Foreign Key Manages dari Branch ke Staff
ALTER TABLE Branch
    ADD CONSTRAINT Branch_MgrStaff_FK FOREIGN KEY (mgrStaffNo) REFERENCES Staff (staffNo);

-- 3. Tabel Manager
CREATE TABLE Manager (
    staffNo VARCHAR(10) NOT NULL,
    mgrStartDate DATE,
    bonus NUMERIC(10, 2),
    CONSTRAINT Manager_PK PRIMARY KEY (staffNo),
    CONSTRAINT Manager_Staff_FK FOREIGN KEY (staffNo) REFERENCES Staff (staffNo)
);

-- 4. Tabel Telephone
CREATE TABLE Telephone (
    telNo VARCHAR(20) NOT NULL,
    branchNo VARCHAR(10) NOT NULL,
    CONSTRAINT Telephone_PK PRIMARY KEY (telNo),
    CONSTRAINT Telephone_Branch_FK FOREIGN KEY (branchNo) REFERENCES Branch (branchNo)
);

-- 5. Tabel Client
CREATE TABLE Client (
    clientNo VARCHAR(10) NOT NULL,
    fName VARCHAR(30),
    lName VARCHAR(30),
    telNo VARCHAR(20),
    prefType VARCHAR(20),
    maxRent NUMERIC(10, 2),
    eMail VARCHAR(50),
    CONSTRAINT Client_PK PRIMARY KEY (clientNo)
);
```

```sql
-- 6. Tabel Registration
CREATE TABLE Registration (
    clientNo VARCHAR(10) NOT NULL,
    branchNo VARCHAR(10) NOT NULL,
    staffNo VARCHAR(10) NOT NULL,
    dateJoined DATE,
    CONSTRAINT Registration_PK PRIMARY KEY (clientNo),
    CONSTRAINT Registration_Client_FK FOREIGN KEY (clientNo) REFERENCES Client (clientNo),
    CONSTRAINT Registration_Branch_FK FOREIGN KEY (branchNo) REFERENCES Branch (branchNo),
    CONSTRAINT Registration_Staff_FK FOREIGN KEY (staffNo) REFERENCES Staff (staffNo)
);

-- 7. Tabel PrivateOwner (POwns PropertyForRent)
CREATE TABLE PrivateOwner (
    ownerNo VARCHAR(10) NOT NULL,
    fName VARCHAR(30),
    lName VARCHAR(30),
    address VARCHAR(100),
    telNo VARCHAR(20),
    eMail VARCHAR(50),
    CONSTRAINT PrivateOwner_PK PRIMARY KEY (ownerNo)
);

-- 8. Tabel BusinessOwner (BOwns PropertyForRent)
CREATE TABLE BusinessOwner (
    ownerNo VARCHAR(10) NOT NULL,
    bName VARCHAR(50),
    bType VARCHAR(30),
    address VARCHAR(100),
    telNo VARCHAR(20),
    eMail VARCHAR(50),
    contactName VARCHAR(50),
    CONSTRAINT BusinessOwner_PK PRIMARY KEY (ownerNo)
);

-- 9. Tabel PropertyForRent (Oversees Staff, Offers Branch, POwns/BOwns Owner)
CREATE TABLE PropertyForRent (
    propertyNo VARCHAR(10) NOT NULL,
    ownerNo VARCHAR(10) NOT NULL,
    staffNo VARCHAR(10),
    branchNo VARCHAR(10) NOT NULL,
    street VARCHAR(50),
    city VARCHAR(30),
    postcode VARCHAR(15),
    type VARCHAR(20),
    rooms INTEGER,
    rent NUMERIC(10, 2),
    CONSTRAINT PropertyForRent_PK PRIMARY KEY (propertyNo),
    CONSTRAINT Property_Staff_FK FOREIGN KEY (staffNo) REFERENCES Staff (staffNo),
    CONSTRAINT Property_Branch_FK FOREIGN KEY (branchNo) REFERENCES Branch (branchNo),
    CONSTRAINT Property_PrivateOwner_FK FOREIGN KEY (ownerNo) REFERENCES PrivateOwner (ownerNo),
    CONSTRAINT Property_BusinessOwner_FK FOREIGN KEY (ownerNo) REFERENCES BusinessOwner (ownerNo)
);
```

-

```sql
-- 10. Tabel Lease
CREATE TABLE Lease (
    leaseNo VARCHAR(10) NOT NULL,
    clientNo VARCHAR(10) NOT NULL,
    propertyNo VARCHAR(10) NOT NULL,
    rent NUMERIC(10, 2),
    paymentMethod VARCHAR(20),
    deposit NUMERIC(10, 2),
    paid CHAR(1),
    rentStart DATE,
    rentFinish DATE,
    CONSTRAINT Lease_PK PRIMARY KEY (leaseNo),
    CONSTRAINT Lease_Client_FK FOREIGN KEY (clientNo) REFERENCES Client (clientNo),
    CONSTRAINT Lease_Property_FK FOREIGN KEY (propertyNo) REFERENCES PropertyForRent (propertyNo)
);

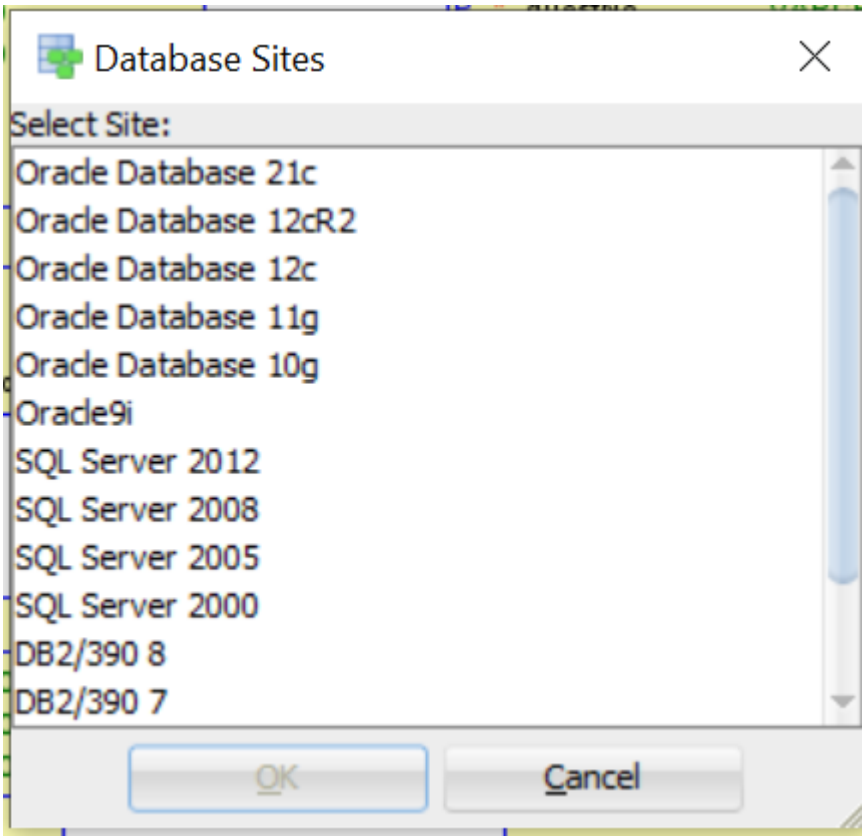
-- 11. Tabel Viewing
CREATE TABLE Viewing (
    clientNo VARCHAR(10) NOT NULL,
    propertyNo VARCHAR(10) NOT NULL,
    viewDate DATE NOT NULL,
    comment VARCHAR(200),
    CONSTRAINT Viewing_PK PRIMARY KEY (clientNo, propertyNo),
    CONSTRAINT Viewing_Client_FK FOREIGN KEY (clientNo) REFERENCES Client (clientNo),
    CONSTRAINT Viewing_Property_FK FOREIGN KEY (propertyNo) REFERENCES PropertyForRent (propertyNo)
);

-- 12. Tabel Newspaper
CREATE TABLE Newspaper (
    newspaperName VARCHAR(50) NOT NULL,
    address VARCHAR(100),
    telNo VARCHAR(20),
    contactName VARCHAR(50),
    CONSTRAINT Newspaper_PK PRIMARY KEY (newspaperName)
);

-- 13. Tabel Advert
CREATE TABLE Advert (
    propertyNo VARCHAR(10) NOT NULL,
    newspaperName VARCHAR(50) NOT NULL,
    dateAdvert DATE NOT NULL,
    cost NUMERIC(10, 2),
    CONSTRAINT Advert_PK PRIMARY KEY (propertyNo, newspaperName, dateAdvert),
    CONSTRAINT Advert_Property_FK FOREIGN KEY (propertyNo) REFERENCES PropertyForRent (propertyNo),
    CONSTRAINT Advert_Newspaper_FK FOREIGN KEY (newspaperName) REFERENCES Newspaper (newspaperName)
);
```

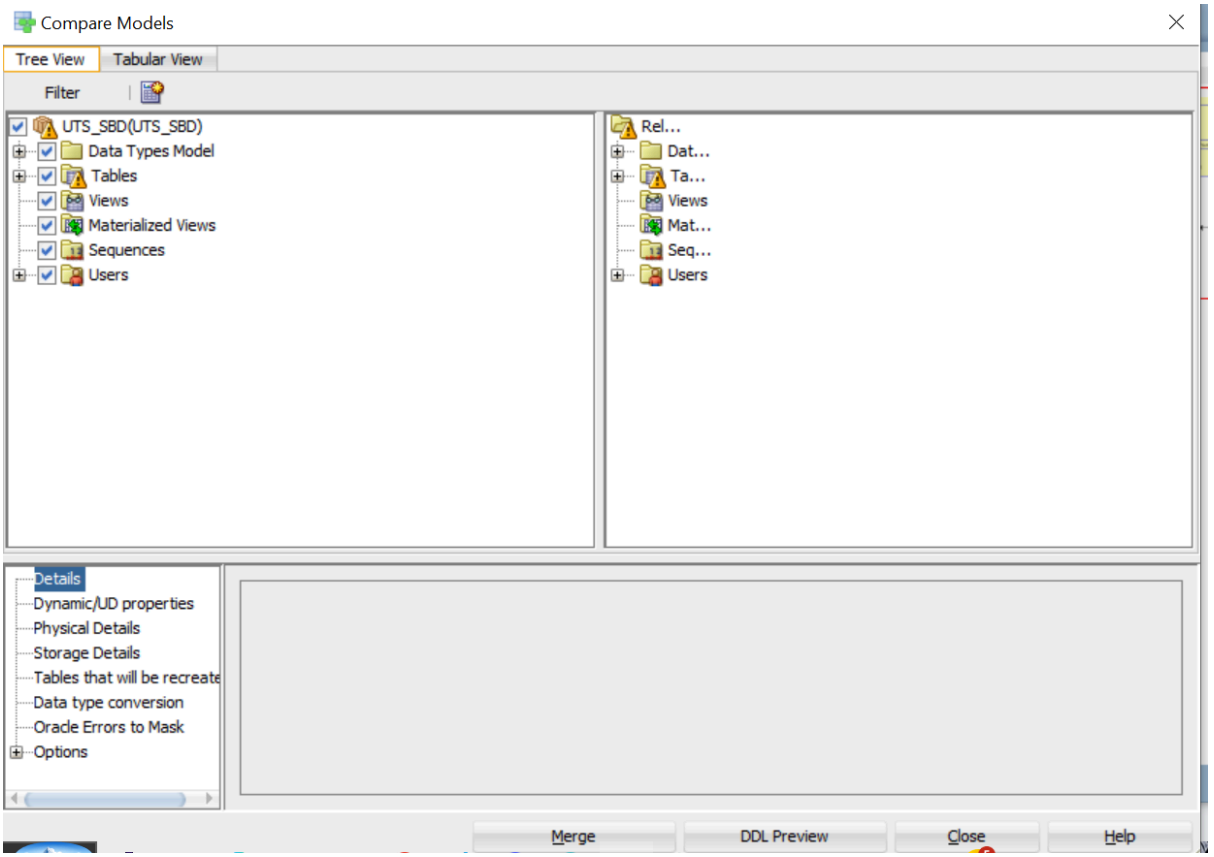
-

3. Lalu import file tersebut ke Oracle

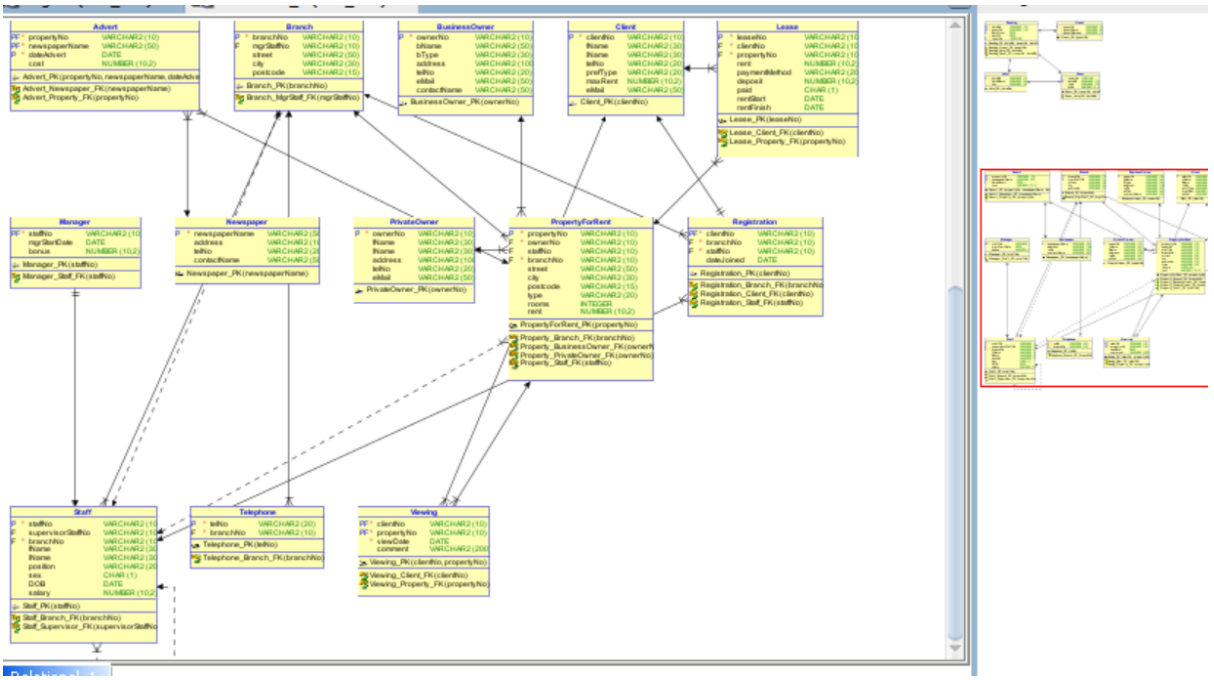


   - Pada bagian ini pilih pada oracle 12c

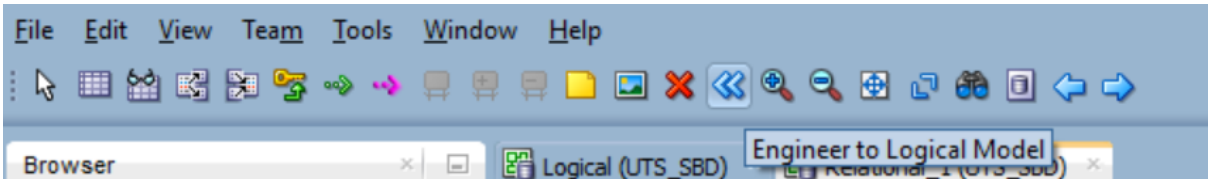
4. Simpan query di notepad sebagai .sql
5. Import ke Data Modeler.
6. Buat physical model dengan target PostgreSQL.
7. Generate DDL dari physical model
8. Lalu Kita merge



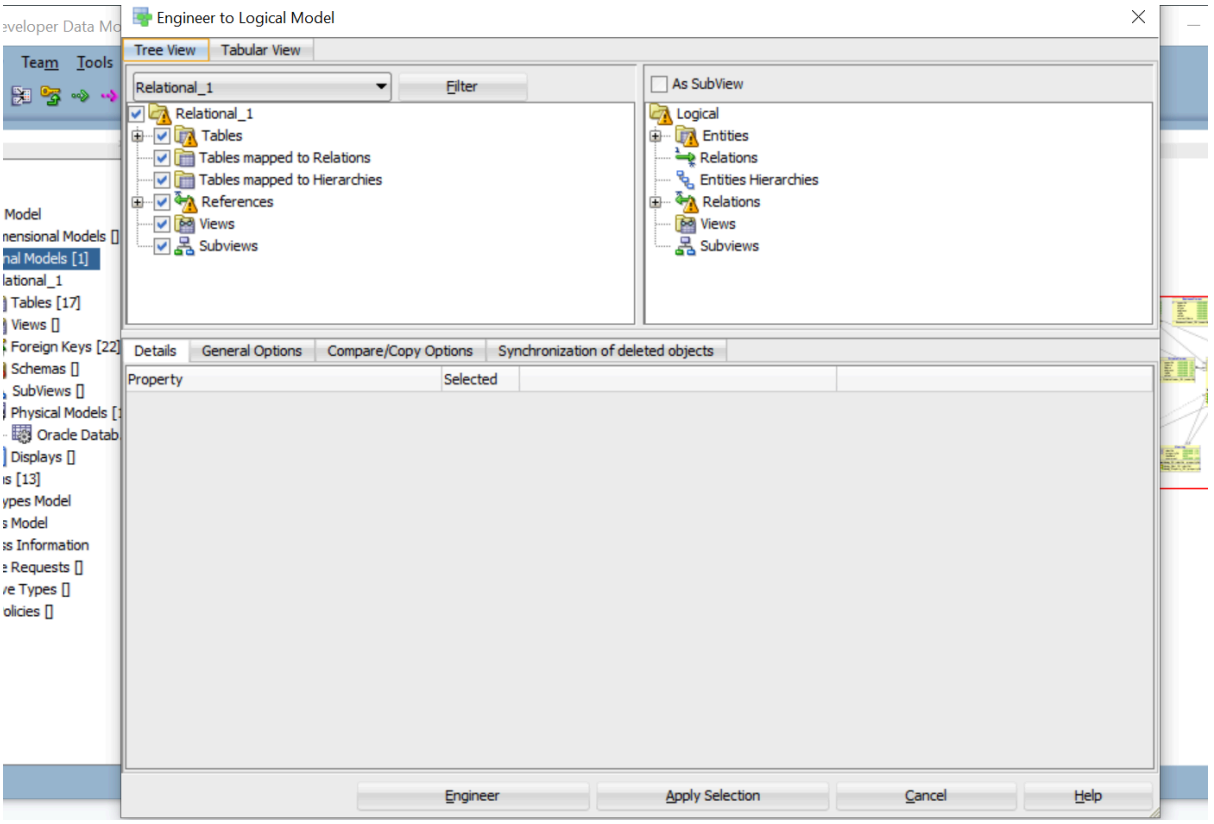
9. Maka akan menghasilkan relasi tersebut



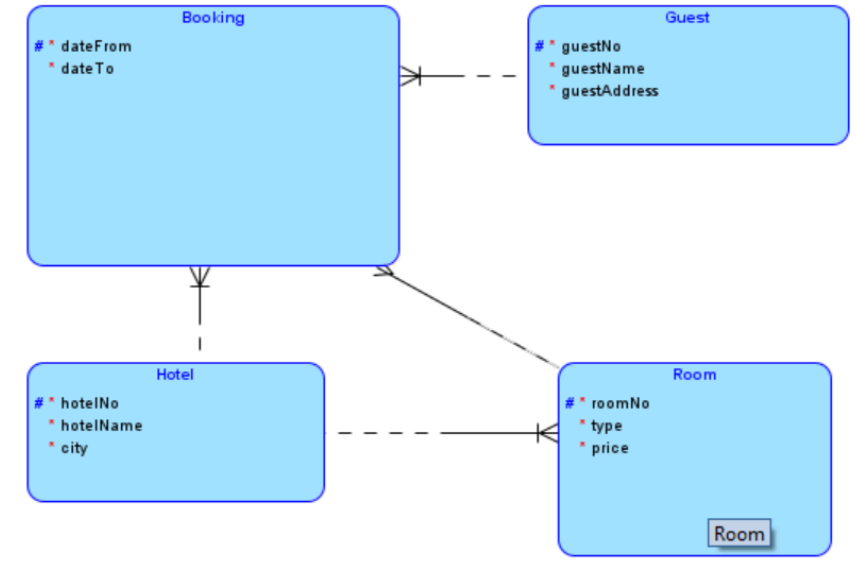
10. Langkah selanjutnya membuat phsyical model pada oracle
    Ikuti arahann tersebut pada gambar ikon ikon panah



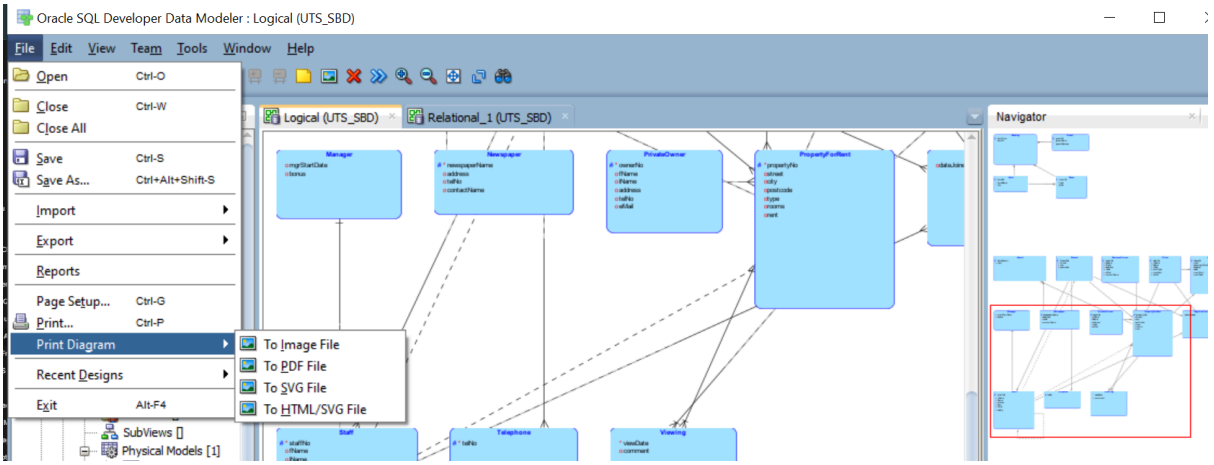
11. Setelah itu klik pada bagian Enggineer model
    - Klik Enggineer



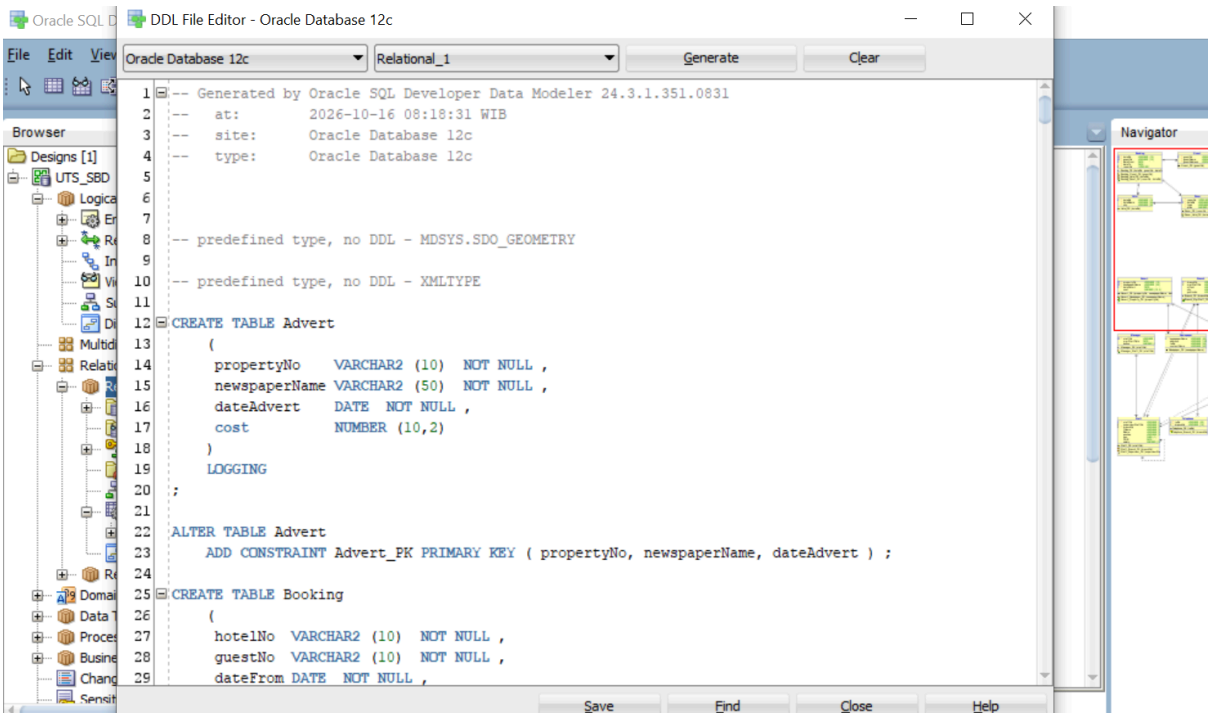
12. Langkah selanjutnya membuat Logical Model (ERD berwarna biru) dengan menekan ikon panah biru (*Engineer to Logical Model*).
    seperti ini mulai dari sql hotel tadi, dan dreamrooms



13. Setelah jadi logic phsicalnya kita save aja ke dalam image



GENERATE TABLE DDL




```
-- Generated by Oracle SQL Developer Data Modeler 24.3.1.351.0831
--   at:        2026-10-16 08:18:31 WIB
--   site:      Oracle Database 12c
--   type:      Oracle Database 12c



-- predefined type, no DDL - MDSYS.SDO_GEOMETRY

-- predefined type, no DDL - XMLTYPE

CREATE TABLE Advert
    (
     propertyNo    VARCHAR2 (10)  NOT NULL ,
     newspaperName VARCHAR2 (50)  NOT NULL ,
     dateAdvert    DATE  NOT NULL ,
     cost          NUMBER (10,2)
    )
    LOGGING
;

ALTER TABLE Advert
    ADD CONSTRAINT Advert_PK PRIMARY KEY ( propertyNo, newspaperName, dateAdvert ) ;

CREATE TABLE Booking
    (
     hotelNo  VARCHAR2 (10)  NOT NULL ,
     guestNo  VARCHAR2 (10)  NOT NULL ,
     dateFrom DATE  NOT NULL ,
```

DDL File Editor - Oracle Database 12c

Oracle Database 12c | Relational_1 | Generate | Clear

```
ALTER TABLE Booking
    ADD CONSTRAINT Booking_PK PRIMARY KEY ( hotelNo, guestNo, dateFrom ) ;

CREATE TABLE Branch
    (
     branchNo   VARCHAR2 (10)  NOT NULL ,
     mgrStaffNo VARCHAR2 (10) ,
     street     VARCHAR2 (50) ,
     city       VARCHAR2 (30) ,
     postcode   VARCHAR2 (15)
    )
    LOGGING
;

ALTER TABLE Branch
    ADD CONSTRAINT Branch_PK PRIMARY KEY ( branchNo ) ;

CREATE TABLE BusinessOwner
    (
     ownerNo     VARCHAR2 (10)  NOT NULL ,
     bName       VARCHAR2 (50) ,
     bType       VARCHAR2 (30) ,
     address     VARCHAR2 (100) ,
     telNo       VARCHAR2 (20) ,
     eMail       VARCHAR2 (50) ,
     contactName VARCHAR2 (50)
    )
    LOGGING
```

```
CREATE TABLE Client
    (
     clientNo VARCHAR2 (10)  NOT NULL ,
     fName    VARCHAR2 (30) ,
     lName    VARCHAR2 (30) ,
     telNo    VARCHAR2 (20) ,
     prefType VARCHAR2 (20) ,
     maxRent  NUMBER (10,2) ,
     eMail    VARCHAR2 (50)
    )
    LOGGING
;

ALTER TABLE Client
    ADD CONSTRAINT Client_PK PRIMARY KEY ( clientNo ) ;

CREATE TABLE Guest
    (
     guestNo      VARCHAR2 (10)  NOT NULL ,
     guestName    VARCHAR2 (50)  NOT NULL ,
     guestAddress VARCHAR2 (100)  NOT NULL
    )
    LOGGING
;

ALTER TABLE Guest
    ADD CONSTRAINT Guest_PK PRIMARY KEY ( guestNo ) ;

CREATE TABLE Hotel
    (
     hotelNo   VARCHAR2 (10)  NOT NULL ,
     hotelName VARCHAR2 (50)  NOT NULL ,
     city      VARCHAR2 (50)  NOT NULL
    )
    LOGGING
;

ALTER TABLE Hotel
    ADD CONSTRAINT Hotel_PK PRIMARY KEY ( hotelNo ) ;

CREATE TABLE Lease
    (
     leaseNo       VARCHAR2 (10)  NOT NULL ,
     clientNo      VARCHAR2 (10)  NOT NULL ,
     propertyNo    VARCHAR2 (10)  NOT NULL ,
     rent          NUMBER (10,2) ,
     paymentMethod VARCHAR2 (20) ,
     deposit       NUMBER (10,2) ,
     paid          CHAR (1) ,
     rentStart     DATE ,
     rentFinish    DATE
    )
    LOGGING
```

PENDING GITHUB

